

Lirui Wang

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EDUCATION

University of Washington, Seattle, WA

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

GPA: 3.97 / 4.00, Advisor: Prof. Dieter Fox

Sep. 2016 – Jun. 2020

University of Washington, Seattle, WA

Master of Science, Computer Science, College of Arts and Sciences

Sep. 2020 – Jun. 2021

RESEARCH PROJECTS

Motion and Grasp Planning, Seattle, WA

UW Robotics and States Estimation Lab

March. 2019 – Present.

- Implement a novel approach that combines data-driven and traditional method to synthesize grasp pose
- Prototype RL, IL, and MPC with PyBullet environment to map observed points to 6D pose increments in task space
- Research extensively on traditional methods and data-driven methods in grasp synthesis
- Jointly optimize motion and grasp planning in cluttered scene by extending the CHOMP and online learning frameworks

DeepIM Tracking, Seattle, WA

UW Robotics and States Estimation Lab

June. 2018 – March. 2019

- Tested extensively DeepIM on YCB datasets and achieved robust results trained on synthetic data only
- Improved network training stability, pose distribution and occlusion generation
- Extended DeepIM in real-time tracking for occluded scene in YCB and built AR demos
- Worked on robot kinematics to build generalized vision system that can track robot arm

Depth and Flow, Seattle, WA

UW Robotics and States Estimation Lab

March. 2018 – Jun. 2018

- Investigated a few deep learning methods results for jointly 3D geometry predictions such as depths, flow, and egomotion
- Explored unsupervised structures and ideas such as SfMLearner and MonoDepth and analyzed them on a robotic setting

Baxter Manipulation, Seattle, WA

UW Robotics and States Estimation Lab

Aug. 2017 – March. 2018

- Worked on a manipulation task with PoseCNN, which enables Baxter to manipulate daily objects
 - Researched on camera models, robot kinematics, 3D geometry and ROS packages for tracking and control
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EXPERIENCE

Teaching Assistant, Seattle, WA

UW CSE 455 (Computer Vision)

Sep. 2019 – Present.

Research Assistant, Seattle, WA

UW Adaptive Computing Machines and Emulator Lab

March. 2018 – Jun. 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle

Computer Vision Lead & Drone Lead, Seattle, WA

Advanced Robotics at the University of Washington

Sep. 2016 – Feb. 2018

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI
- Perform system integration of controls, navigation and serial communication RoboMasters hosted by DJI

Robotics Engineer Intern, Shenzhen, China

DJI Technology

Jun. 2017 – Aug. 2017

- Worked on automatic robot battle of vehicles with TX series onboard and DJI Manifold with Guidance
 - Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
 - Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot
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SKILLS & OTHERS

Programming Language: Java, C++/C, Python, Matlab

Frameworks: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32

Tools: Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt

Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing, Statistical Learning, High-Dimension Statistics, Robust Machine Learning

Scholarships: Tau Beta Pi Scholarship 2017-2019, Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship 2017-2018, Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship 2018-2019